

PHABS - Portable Haptic Assisted Bimanual System

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Abstract—Teleoperation-based data collection is among the most effective methods for training robot manipulation policies, yet existing handheld systems discard force information entirely, limiting both operator performance and data quality. We present PHABS (Portable Haptic Assisted Bimanual System), a handheld bimanual haptic teleoperation device designed for quasi-symmetric two-handed manipulation tasks. The device provides pinching force feedback to both hands via capstan-driven actuators, with passive two-axis gimbal mounts accommodating wrist orientation. A linear capstan drive was designed for lateral force feedback, though binding in the prototype prevented its use in evaluation. Position and force data are captured simultaneously, during operation via potentiometer and encoder sensing, enabling high-quality demonstration collection for robot learning. We evaluated the pinching subsystem through a pilot user study in which participants performed stiffness discrimination and object manipulation tasks, with and without haptic feedback. Results indicate that haptic feedback meaningfully improves operator confidence and task success rate. We discuss lessons learned from the design and fabrication process, and outline directions for future development.

I. INTRODUCTION

Robots are increasingly deployed in manufacturing, healthcare, and logistics to perform tasks that are dangerous, repetitive, or physically demanding. Yet teaching robots to manipulate objects with human-level dexterity remains a persistent challenge. One of the most effective approaches is teleoperation-based data collection: a human operator remotely controls the robot while their movements are recorded as training demonstrations. However, existing handheld teleoperation systems capture only hand position, discarding force information entirely. Without haptic feedback, the operator cannot feel how hard they are gripping, whether an object is slipping, or how much force is needed to interact with the environment. This makes delicate two-handed tasks, such as handling fragile objects and opening sealed packaging, slow, imprecise, and difficult to demonstrate reliably.

The quality of its training data limits the quality of autonomous robot behavior. Poor teleoperation produces unnatural, hesitant demonstrations that fail to capture the nuanced force profiles human manipulation relies on. As the robotics community scales toward larger and more diverse datasets, the need for high-quality, force-rich demonstration data becomes increasingly critical to continue improving outcomes.

We present a handheld bimanual haptic teleoperation device designed to address this gap. By embedding force feedback directly into the device, operators receive continuous

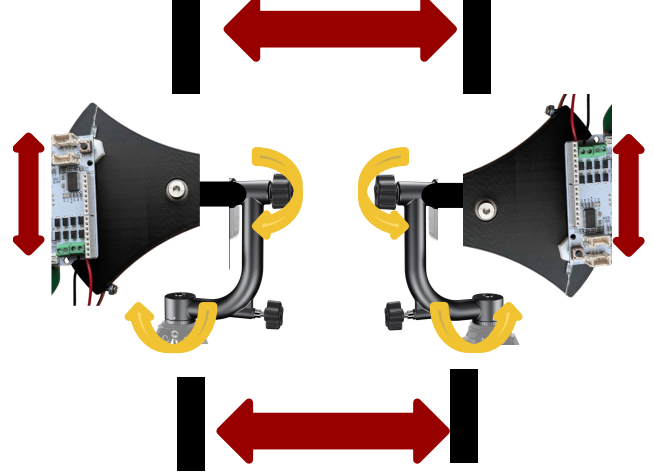


Fig. 1. Conceptual design of the PHABS device, with red indicating actuated degrees of freedom that provide haptic force feedback and yellow indicating passive degrees of freedom used for position sensing only. The design targets bimanual quasi-symmetric manipulation tasks, with the goal of improving both teleoperation performance and the quality of demonstration data collected for robot learning.

tactile information about the robot’s physical interactions, thereby closing the sensorimotor loop on which skilled human manipulation depends. The system is designed for quasi-symmetric bimanual tasks. In these two-handed operations, both hands play active, coordinated roles, a class of tasks underrepresented in existing datasets precisely because they are difficult to capture without haptics. The device is also cross-embodiment compatible, allowing the same operator to collect demonstrations across robot platforms of varying size and strength, including robots capable of exerting forces well beyond human scale.

Beyond teleoperation performance, this approach has broader societal implications. Automation is advancing faster than many workers can retrain for new roles. A system that keeps humans meaningfully in the loop, leveraging existing human skill to guide and train robots, preserves human expertise during this transition and produces better robots in the process.

In this paper, we provide the following technical contributions:

- A handheld bimanual haptic teleoperation device for quasi-symmetric two-handed manipulation
- A dataset collection pipeline capturing simultaneous motion and force data during teleoperation
- A user study evaluating task performance with and without haptic feedback

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II. RELATED WORKS

A. Haptic Teleoperation

Haptic feedback has been shown to improve operator performance in teleoperation tasks meaningfully. Purushottam et al. [1] demonstrate haptic teleoperation of a wheeled humanoid robot performing heavy lifting tasks, evaluating multiple control strategies and pilot preferences. Their findings highlight the importance of carefully mapping forces between the teleoperated robot and the haptic device and show that user preferences for haptic feedback vary with the task phase, for example, preferring neutralized feedback when releasing a load. Lenz and Behnke [2] present a bimanual exoskeleton system that provides force and haptic feedback to the operator's arms and hands during teleoperation of an anthropomorphic robot avatar. Their system uses force-torque sensors at the robot's end effectors to relay contact forces to the user. It introduces an oscillation observer to maintain stability when feedback forces could destabilize the controller. Together, these works establish the viability of force-feedback teleoperation for complex manipulation and inform our approach to bimanual force rendering and stability.

B. Shared Control and Dexterous Manipulation

Griffin et al. [3] examine feedback strategies for shared control in dexterous telemanipulation, demonstrating that keeping a human in the loop with appropriate tactile, force, and motion feedback significantly improves task performance. Their work motivates our use of haptic feedback not only for teleoperation quality but also to generate higher-quality training demonstrations. Milstein et al. [4] investigate grip force control during interaction with both rigid and deformable objects via a haptic gripper, finding that haptic properties of the gripper, rather than visual feedback alone, govern the forces users exert. This provides validation for using haptics for our targeted tasks.

C. Wearable and Handheld Haptic Devices

Jo et al. [5] present WriMouCon, a wrist-mounted haptic controller that renders physical properties such as stiffness, viscosity, and object weight in virtual reality. Their analysis of the minimum perceivable force differences informs our design choices for force rendering resolution in our handheld device. In contrast to exoskeleton-based systems [2], our device prioritizes portability and cross-embodiment compatibility, enabling deployment across different robot platforms without requiring the operator to wear a full upper-body apparatus.

D. Teleoperation for Robot Learning

Mobile ALOHA [6] is among the closest existing systems to our work, combining a mobile base with bimanual teleoperation for whole-body data collection and imitation learning. However, Mobile ALOHA lacks haptic feedback and force sensing, and its demonstrations are tied to its own hardware

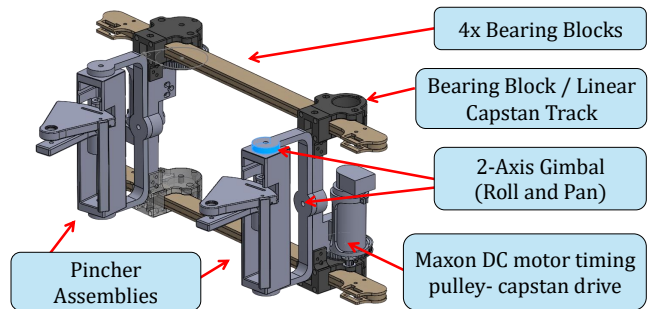


Fig. 2. Full mechanical assembly of the PHABS device, showing the two pincher assemblies mounted on bearing block carriages that translate along parallel wooden rails. The central column houses the Maxon DC motor and capstan drive for lateral force feedback, with two-axis gimbals connecting each pincher to its respective carriage.

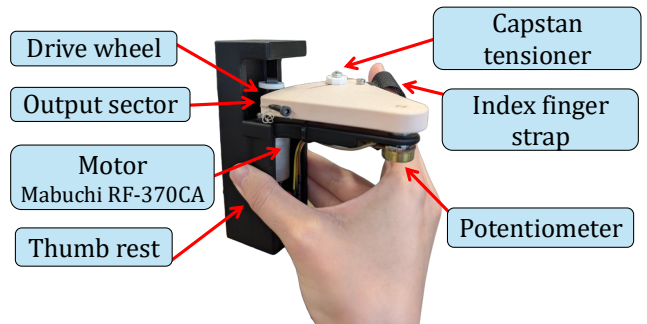


Fig. 3. Pinch feedback mechanism in right-handed configuration. The Mabuchi RF-370CA motor drives the output sector via a direct-drive capstan. The user's thumb rests on the housing while the index finger is strapped to the sector, with the potentiometer serving as both the sector pivot and angle sensor.

platform. Our system addresses both limitations by incorporating force feedback and supporting cross-embodiment deployment.

Collectively, this body of work demonstrates that haptic feedback improves both teleoperation performance and operator experience, particularly for tasks involving deformable objects and fine force control. However, existing systems are either platform-specific, limited to unimanual operation, or unable to capture force and motion data for robot learning simultaneously. Our work addresses these gaps with a handheld, bimanual, cross-embodiment haptic teleoperation device designed explicitly for dataset generation.

III. MECHANICAL DESIGN

The device was designed to provide two modalities of haptic force feedback: lateral force feedback for tasks such as squeezing a box from both sides, and pinching force feedback between the thumb and index finger, as shown in Fig. 2. Each pincher assembly was mounted onto a two-axis gimbal with a pan axis (rotation about the vertical axis of the housing) and a roll axis (rotation about the longitudinal axis of the housing).

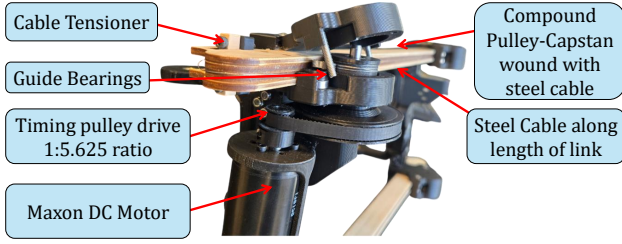


Fig. 4. Linear capstan drive assembly for lateral force feedback, showing the Maxon DC motor, 1:5.625 timing pulley stage, compound pulley-capstan wound with steel cable, guide bearings, and cable tensioner. The steel cable transmits motor torque along the length of the output link to the bearing block carriage.

A. Pincher

The pinching force feedback mechanism was designed based on the Force Dimension omega.7 end-effector [7] to fit the natural shape and range of motion of the hand while grasping, providing force feedback between the thumb and index finger, as shown in Fig. 3.

The mechanism is actuated via a direct-drive capstan, in which a Mabuchi RF-370CA motor rotates a drive wheel that winds a cable around the output sector. The sector rotates about a potentiometer, which serves as both the pivot point and the angle sensor. A capstan tension adjustment screw maintains cable preload to minimize backlash. During operation, the user rests their thumb on the thumb rest on the housing, while their index finger is strapped to a handle on the output sector, allowing active force feedback during both opening and closing of the hand. The remaining fingers grasp the housing to provide stability. The housing has mirrored configurations for the left and right hands.

The output pinching force is given by:

$$F_{pinch} = \frac{r_s}{r_h r_d} \tau_{motor} \quad (1)$$

where $\tau_{motor} = 2.48 \text{ mN} \cdot \text{m}$ is the torque output of the Mabuchi RF-370CA motor, $r_s = 75 \text{ mm}$ is the radius of the output sector, $r_h = 60 \text{ mm}$ is the radius of the handle, and $r_d = 4.75 \text{ mm}$ is the radius of the drive wheel. The ratio r_s/r_d captures the mechanical advantage of the capstan transmission between the drive wheel and the output sector. With these specifications, the maximum expected output force is $F_{pinch} = 0.65 \text{ N}$.

The pincher assembly is mounted to the lateral mechanism via a two-axis gimbal. The first axis is aligned with the longitudinal axis of the housing (roll), accommodating rotation of the wrist about the forearm axis. The second axis is perpendicular to the housing face (pan), accommodating lateral wrist flexion. Together, these two passive rotational degrees of freedom allow the device to passively align with an arbitrarily oriented target object before grasping. This also enables manipulation of objects already held in the hand, supporting tasks such as passing objects between hands, opening packaging, and tightening screws.

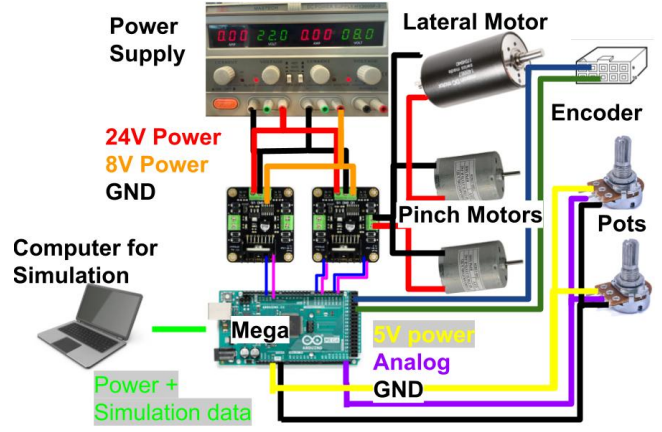


Fig. 5. Wiring diagram for the PHABS device. The Arduino Mega serves as the central controller, receiving power from the simulation computer via USB and relaying sensor data (potentiometer angles and encoder counts) to the simulator while receiving force commands in return. The dual-rail power supply provides 22V to the three motors and 8V to the motor controller logic.

B. Lateral Force Feedback

The lateral force feedback mechanism uses a linear capstan drive, based on the design of a dynamically extensible leg mechanism [8], consisting of two parallel wooden rails along which two bearing block carriages translate, as shown in Fig. 2. Each carriage consists of a pair of bearing blocks clamping onto the rail from above and below, with ball bearings contacting the rail surface to allow for smooth, low-friction sliding. The bearing blocks feature adjustable screws, allowing the contact force of the bearings against the rail to be tuned during assembly to eliminate play without introducing binding from external moments acting on the carriage.

One carriage is actuated by the Maxon DC motor via the capstan drive, shown in Fig. 4, to render resistance forces, while the opposing carriage is fixed in place, providing a static reaction surface for the operator to push against. Each carriage mounts a gimbal and pincher assembly, so that when the operator squeezes their hands together, the actuated carriage displaces along the rails and the motor applies a programmable resistive force corresponding to the simulated environment.

IV. HARDWARE

The system is controlled by an Arduino Mega, which handles all sensor data acquisition and motor control as seen in Fig. 5, communicating with the simulation computer via serial connection at 38,400 baud. Due to the length of the cable runs between the device and the simulation computer, higher baud rates introduced significant line noise and communication errors; 38,400 baud was found to be the highest reliable rate for the wiring lengths used.

A. Sensing

Each pincher is equipped with a 10K potentiometer that measures the angular position of the pincher jaw. The raw

ADC counts are mapped to a linear chord distance in millimeters using:

$$d = \frac{N_{\text{closed}} - N}{N_{\text{closed}} - N_{\text{open}}} \cdot d_{\text{max}} \quad (2)$$

where N is the raw ADC reading, N_{closed} and N_{open} are the calibrated counts at the fully closed and fully open positions, respectively, and $d_{\text{max}} = 20$ mm is the maximum jaw opening. The right and left pinchers use calibrated ranges of $[430, 600]$ and $[450, 590]$ counts respectively. To improve robustness, readings outside the valid calibration range are rejected and the last valid measurement is held. An IIR low-pass filter is applied to each potentiometer signal before transmission:

$$\hat{d}[n] = (1 - \alpha) \hat{d}[n - 1] + \alpha d[n] \quad (3)$$

with $\alpha = 0.05$ for the left pincher and $\alpha = 0.10$ for the right pincher.

The Maxon motor's quadrature encoder (500 counts per revolution) tracks lateral slider displacement. The encoder count is converted to linear distance as:

$$d_{\text{slider}} = \frac{N_{\text{enc}}}{4 \cdot \text{CPR}} \cdot 2\pi r \cdot G \quad (4)$$

where $\text{CPR} = 500$, $r = 12.5$ mm is the capstan drive wheel radius, and $G = 2.625$ is the mechanical gain of the drive. The factor of 4 accounts for quadrature decoding on both rising and falling edges of both encoder channels. Slider displacement is clamped to $[0, 400]$ mm and filtered with $\alpha = 0.05$.

B. Motor Control and Communication

Force commands are received from the simulation computer as a 14-byte binary packet framed by header byte $0 \times \text{BB}$ and footer byte $0 \times \text{FF}$, containing three 4-byte floats representing the commanded forces for the slider, left pincher, and right pincher, respectively. These force values, expressed as a percentage of maximum output, are converted to PWM duty cycles as:

$$\text{PWM} = \left\lfloor \frac{F}{100} \cdot 255 \right\rfloor \quad (5)$$

Sensor data is transmitted back to the simulation computer in the same binary framing format, containing the filtered slider displacement and both pincher distances.

Power is provided by a dual-rail supply: the 22V rail drives all three motors, and the 8V rail powers the motor controller logic. Although the Mabuchi motors are nominally rated for 12V, they were operated on the 22V rail at a reduced PWM duty cycle to remain within safe thermal limits, with motor temperature monitored during operation. The potentiometers and encoder are powered from the Arduino's 5V rail, and the Arduino itself is powered over USB by the simulation computer.

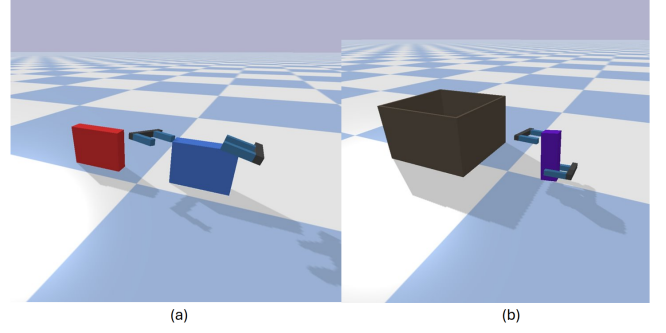


Fig. 6. (a) Virtual environment for the stiffness comparison task. (b) Virtual environment for the block movement task.

V. SOFTWARE

To provide haptic feedback during pinching interactions, objects in the virtual environment were modeled with a stiffness and damping coefficient. The force rendered to the user was given by $F = kx + b\dot{q}$, where x is the penetration depth of the gripper finger into the object surface and $\dot{q}$ is the filtered finger closing velocity.

The virtual environment was rendered using PyBullet. Two HTC Vive trackers were used to capture the position and orientation of each hand, and finger aperture was read from the Arduino haptic device via serial communication. Haptic feedback was implemented using a god-object proxy model. Upon contact with an object, the finger joint of the URDF was clamped at the contact surface to prevent visual penetration, and the difference between the commanded and proxy positions was used to compute the penetration depth for force rendering.

The URDF gripper model consisted of two C-shaped grippers, each comprising a rigid spine with a fixed finger and a sliding finger connected via a prismatic joint. Each gripper was positioned and oriented in the simulation by a kinematic chain of three prismatic joints and three continuous rotation joints, driven directly by the position and orientation of its corresponding Vive tracker. A third Vive tracker was used to define a shared coordinate frame between the physical workspace and the PyBullet simulation, ensuring accurate mapping from user motion to gripper motion.

Two experimental scenes were developed, as seen in Fig. 6. In the first scene, users interacted with two blocks of differing stiffness ($k = 2000$ N/m and $k = 1800$ N/m), allowing them to discriminate between materials through haptic feedback alone. In the second scene, users were presented with a fragile object and an open bin. The task required picking up the object with the right hand, transferring it to the left hand, and placing it in the bin. If the object was squeezed beyond a force threshold, it was deleted from the scene, simulating breakage and incentivizing careful handling to evaluate whether haptic feedback could support delicate manipulation.

VI. METHODS

Once we had completed assembly of the device, a few issues were found with the lateral force feedback and gimbal

Algorithm 1 Main Simulation Loop

```

1: while simulation is running do
2:   Read sensor inputs
3:    $q_L, q_R \leftarrow$  pincher position from potentiometers
4:    $\mathbf{p}_L, \mathbf{p}_R, \mathbf{R}_L, \mathbf{R}_R \leftarrow$  position and orientation from
     Vive trackers
5:
6:   Update URDF gripper poses
7:   Drive prismatic joints to  $\mathbf{p}_L, \mathbf{p}_R$ 
8:   Drive rotation joints to  $\mathbf{R}_L, \mathbf{R}_R$ 
9:
10:  Compute filtered finger velocity
11:   $\hat{q}_g = \alpha (q_g - q_g^{\text{prev}})/dt + (1 - \alpha) \hat{q}_g$  for  $g \in \{L, R\}$ 
12:
13:  Determine proxy finger position
14:  for each gripper  $g \in \{L, R\}$  do
15:    if first contact frame then
16:       $q_g^* = q_g$   $\triangleright$  Snapshot surface position
17:    else if no contact then
18:       $q_g^* = \emptyset$ 
19:    end if
20:    if  $q_g^* \neq \emptyset$  and  $q_g < q_g^*$  then
21:      Drive finger joint to  $q_g^*$   $\triangleright$  Clamp at surface
22:    else
23:      Drive finger joint to  $q_g$   $\triangleright$  Free motion
24:    end if
25:  end for
26:
27:  Calculate and render haptic force
28:  for each gripper  $g \in \{L, R\}$  do
29:    if  $q_g^* \neq \emptyset$  then
30:       $x_g = q_g^* - q_g$   $\triangleright$  Penetration depth
31:       $F_g = \min(k x_g + b \hat{q}_g, F_{\max})$ 
32:    else
33:       $F_g = 0$ 
34:    end if
35:  end for
36:  Transmit  $(F_L, F_R)$  to motor controller
37: end while

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mechanisms. This led to the decision to pivot to using only the pinchers and position tracking in 3D space, to allow users to pick up and interact with objects of varying stiffnesses.

A pilot user study was conducted to evaluate the effect of haptic feedback on operator performance during two-handed virtual object manipulation. Participants were seated in front of a monitor displaying the PyBullet simulation environment, holding one pincher device in each hand as seen in Fig. 7. HTC Vive trackers were attached to each hand to provide 6-DOF position and orientation tracking, with four base stations positioned around the participant to ensure full coverage and minimize occlusion.

Each participant completed two tasks. In the first task, participants were presented with two virtual blocks of differing stiffness and asked to identify the stiffer block by pinching

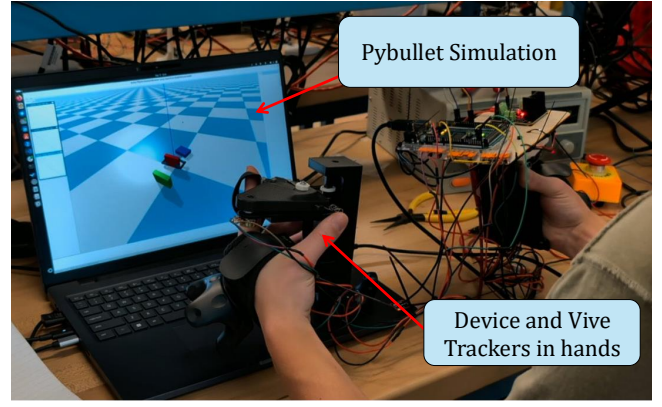


Fig. 7. PHABS in operation during a bimanual virtual object manipulation task. The operator holds one pincher device in each hand, with HTC Vive trackers providing 6-DOF pose to the PyBullet simulation. Capstan-driven force feedback is rendered to both hands in response to virtual contact forces, closing the sensorimotor loop during teleoperation.

each one. In the second task, participants were asked to pick up a target block with one hand, transfer it to the other hand, and place it into a target box. Each task was performed twice: once with haptic feedback disabled and once with it enabled, allowing within-subject comparison of performance. A trial was considered successful if the participant correctly identified the stiffer block in the first task, or completed the full pickup, handoff, and placement sequence without dropping the object in the second. Following the trials, participants were asked to provide qualitative feedback on the perceived usefulness of the haptic feedback and the comfort of the device.

VII. RESULTS

A total of 3 participants completed the pilot user study. In the stiffness discrimination task, 3 of 3 participants correctly identified the stiffer block when haptic feedback was enabled. In the manipulation task, 3 of 3 participants completed the full pickup, handoff, and placement sequence with haptic feedback enabled, compared to 1 of 3 without feedback.

TABLE I
PILOT STUDY RESULTS

Task	Condition	Successful Participants	Total Participants
Stiffness Discrimination	With Haptics	3	3
	Without Haptics	1	3
Manipulation Task	With Haptics	3	3
	Without Haptics	1	3

Qualitative feedback was consistent with the quantitative results. Participants reported that without haptic feedback, it was difficult to determine when contact had been made with a virtual object, leading to hesitant and repeated grasping attempts. With feedback enabled, participants reported greater confidence during grasping and handoff interactions, noting that the force cues allowed them to gauge grip strength in real time. Two of three participants rated the haptic feedback as useful or very useful on a post-trial questionnaire.

VIII. DISCUSSION

The lateral force feedback mechanism was ultimately not suitable for user testing due to excessive binding in the linear capstan drive. The bearing blocks were designed with adjustable preload screws to tune friction, but this introduced a fundamental tradeoff: tightening the screws reduced slop but allowed 3D-printed surfaces to contact the wooden rails; loosening them introduced enough play that moments from the pincher assemblies caused binding. Additional bearings added to the block faces were insufficient to counteract the moment arm introduced by the pincher placement and overall system weight.

The gimbal configuration compounded these issues. When both roll axes align, the system reaches a singularity in which it can no longer support its own weight, causing the device to hang downward. Future iterations should use a lighter, more compact motor placed closer to the capstan drive, and component-level testing should be conducted incrementally to catch design flaws before full integration.

The pinch mechanism's press-fit between the output sector and potentiometer may loosen over time and is not robust against out-of-plane forces. Decoupling the rotation shaft from the sensor in future iterations would reduce friction and improve force transmission. The housing was also sized to the team's hand dimensions, which proved unrepresentative of the broader user population. Future versions should incorporate an adjustable handle and a per-user calibration sequence for the pinch range of motion.

Several hardware improvements would also be prioritized. The current linear PWM mapping does not account for motor nonlinearities such as stiction and back-EMF; empirical motor characterization, as done in HapKit calibration, would produce more accurate force output. Current limiting should be added to prevent motor burnout, as one pincher motor was weakened during testing and degraded feedback fidelity. Finally, shielded cables and proper strain relief would improve signal integrity and allow higher baud rates.

IX. CONCLUSIONS

This work presented PHABS, a portable handheld bimanual haptic teleoperation device for quasi-symmetric two-handed manipulation tasks. The pinching subsystem successfully demonstrated force feedback during virtual object interaction, enabling stiffness discrimination and improving bimanual manipulation performance. The lateral force feedback mechanism, while fully designed and fabricated, was not viable due to the binding from the moment of loading and a gimbal singularity. Nevertheless, the pilot study validates the core premise: haptic feedback meaningfully improves teleoperation performance on tasks requiring fine force control, and by extension, the quality of demonstration data collected for robot learning. Future work should address the mechanical limitations identified here and expand the simulation environment to support a wider range of manipulation tasks.

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